Activity

Scenario Based

1) CCTV image contains salt-and-Pepper noise.

Technique Used -> Median Filter.

Reason -> Removes salt and Pepper noise effectively edges.

2) Satellite image corrupted by Gaussian noise

Technique Used -> Gaussian Filter.

Reason -> smooths Gaussian noise and improves image quality.

3) Medical x-ray has random noise but edges must be preserved.

Technique Used -> Bilateral Filter.

Reason -> Reduces noise while preserving fine edges and details.

4) Scanned document has black and white dots.

Technique Used -> Median Filter.

Reason -> Eliminate isolated black and white dots without affecting text.

5) Low-light image shows heavy grain.

Technique Used -> Non-Local Means (NLM) Filter.

Reason -> Removes grainy noise while retaining image details.

6) Foggy road image has very low contrast.

Technique Used -> Histogram Equalization.

Reason -> Enhances contrast, making objects more visible.

7) Underwater image appears dull.

Technique Used -> Contrast Stretching.

Reason -> Expands the intensity range to improve brightness and color visibility.

8) chest x-ray shows poor visibility of internal structures.

Technique Used -> Adaptive Histogram Equalization (CLAHE).

Reason -> Improves local contrast without over-enhancing noise.

9) Night-time traffic image lacks intensity variation

Technique Used -> Histogram Equalization.

Reason -> Increases intensity variation and improves visibility.

10) Grayscale image uses only a narrow intensity range.

Technique Used -> Contrast stretching.

Reason -> Expands the narrow intensity range to improve overall contrast.